Radians and the Unit Circle

Concept

Radians measure angles using arc length divided by radius.

Convert

180° = π radians; degrees × π/180 = radians.

Circle measures

Arc $s=r\theta$; sector area $A=r^2\theta/2$. These formulas need radians.

Unit circle

$(\cos\theta, \sin\theta)$; use reference angles and quadrant signs.

Reciprocals

$\sec=1/\cos$, $\csc=1/\sin$, $\cot=\cos/\sin$; exclude zero denominators.

Example 1

Convert 225° to radians and find its sine and cosine.

Solution

1. $225\pi/180 = 5\pi/4$; reference angle $\pi/4$, quadrant III.

2. $\sin(5\pi/4) = \cos(5\pi/4) = -\sqrt{2}/2$.

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Example 2

For $r = 6$ cm and $\theta = 2\pi/3$, find arc length and sector area.

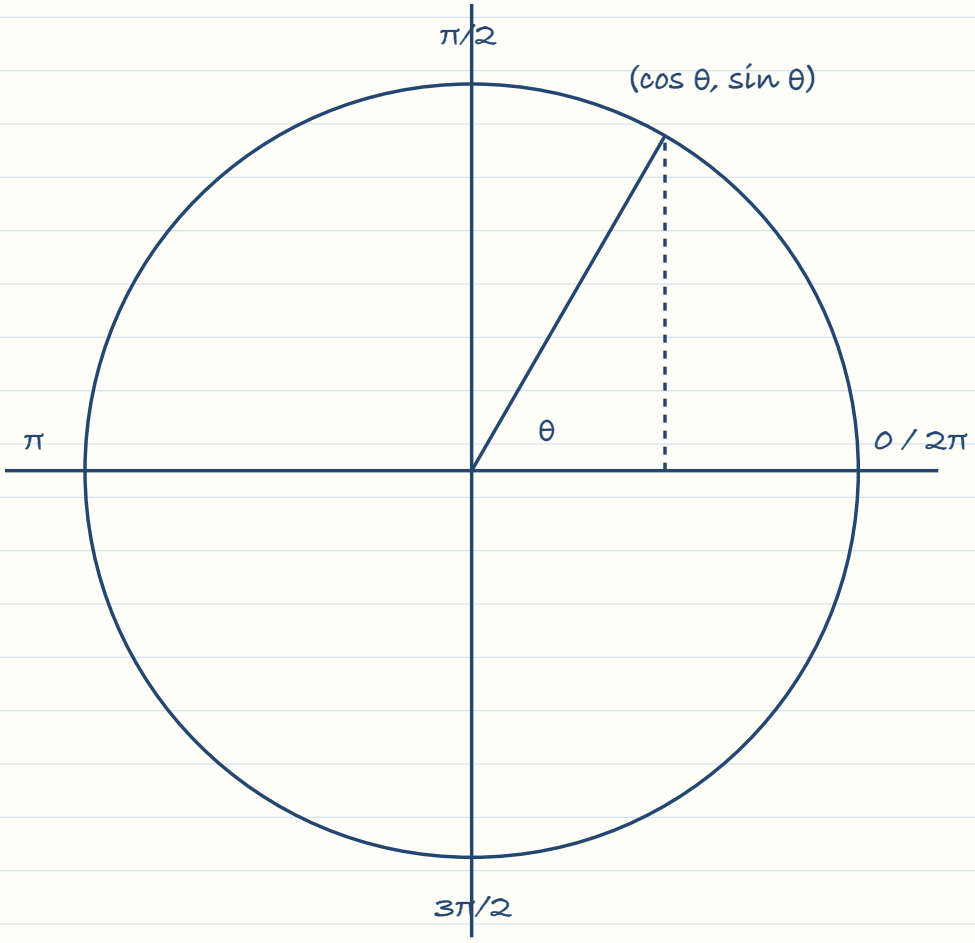
Solution

1. $s = 6(2\pi/3) = 4\pi$ cm.

2. $A = (1/2)(36)(2\pi/3) = 12\pi$ cm^2.

Radians and the Unit Circle

Visual notes | unit circle



The radius is 1. The horizontal coordinate is cosine and the vertical coordinate is sine. The drawn angle is π/3 (60°), giving the exact point (1/2, √3/2).

Signs follow the coordinates: upper half means positive sine; right half means positive cosine. Tangent is sin θ/cos θ and is undefined on the vertical axis.

Radians and the Unit Circle

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Convert $150°$ to radians and $7\pi/6$ to degrees.

2. Find $\sin(5\pi/6)$, $\cos(5\pi/6)$, $\tan(5\pi/6)$.

3. Find the arc length for $r = 8$ m and $\theta = \pi/4$.

4. State where $\tan\theta$ is undefined.

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Practice answers | check after trying

1. $5\pi/6$ and $210°$.

2. $1/2$, $-\sqrt{3}/2$ and $-1/\sqrt{3}$, respectively.

3. $s = 8(\pi/4) = 2\pi$ m.

4. $\theta = \pi/2 + k\pi$, k any integer, where $\cos\theta = 0$.